Exercise Sheet 4

The problems use complex finite-dimensional Lie algebras. The letter κ denotes a Killing form, and $\mathfrak{a}^{\perp}$ is taken with respect to κ. Exercises 4.1–4.3 develop the lecture. Exercises 4.4–4.6 test the limits of its criteria.

Exercise 4.1 (The Killing forms of $\mathfrak{gl}_n$ and $\mathfrak{sl}_n$). Let $n \geq 2$, and let E_{ij} be the matrix units.

(i) For $A, B \in \mathfrak{gl}_n(\mathbb{C})$, prove that the linear map $Z \mapsto AZB$ on $\mathfrak{gl}_n(\mathbb{C})$ has trace $\operatorname{tr} A \operatorname{tr} B$.

(ii) Prove that $\kappa_{\mathfrak{gl}_n}(X, Y) = 2n \operatorname{tr}(XY) - 2 \operatorname{tr} X \operatorname{tr} Y$.

(iii) Deduce $\kappa_{\mathfrak{sl}_n}(X, Y) = 2n \operatorname{tr}(XY)$, and prove that $\mathfrak{sl}_n(\mathbb{C})$ is semisimple. Compare $n = 2$ with the lecture.

(iv) Prove that the radical of $\kappa_{\mathfrak{gl}_n}$ is $\mathbb{C} I_n$.

Exercise 4.2 (Orthogonal ideals and centralisers). Let $\mathfrak{g}$ be semisimple, with simple ideals $\mathfrak{g}_1, \dots, \mathfrak{g}_t$ as in the lecture.

(i) For an ideal $\mathfrak{a}$, prove that $\mathfrak{a}^{\perp} = \{x \in \mathfrak{g} : [x, a] = 0 \text{ for all } a \in \mathfrak{a}\}$.

(ii) Prove that $\kappa(x, y) = 0$ for $x \in \mathfrak{g}_i$, $y \in \mathfrak{g}_j$ and $i \neq j$.

(iii) For Lie algebras $\mathfrak{g}', \mathfrak{g}''$, prove $\kappa_{\mathfrak{g}' \times \mathfrak{g}''}\big((x', x''), (y', y'')\big) = \kappa_{\mathfrak{g}'}(x', y') + \kappa_{\mathfrak{g}''}(x'', y'')$. Deduce that a product of semisimple algebras is semisimple.

(iv) In $\mathfrak{sl}_2 \times \mathfrak{sl}_2$, let $\Delta = \{(x, x)\}$. Prove that $\Delta^{\perp} = \{(x, -x)\}$ and that $\Delta^{\perp}$ is not a subalgebra.

Exercise 4.3 (Semisimple quotients).

(i) For any Lie algebra $\mathfrak{g}$, prove that $\mathfrak{g}/\operatorname{Rad}\mathfrak{g}$ is semisimple.

(ii) Let $\mathfrak{g}$ be semisimple and $\mathfrak{a}$ an ideal. Prove that $\mathfrak{g}/\mathfrak{a} \cong \mathfrak{a}^{\perp}$, and deduce that $\mathfrak{g}/\mathfrak{a}$ is semisimple.

(iii) For $n \geq 2$, prove that $\mathfrak{gl}_n(\mathbb{C})/\mathbb{C} I_n \cong \mathfrak{sl}_n(\mathbb{C})$ and $\operatorname{Rad}\mathfrak{gl}_n(\mathbb{C}) = \mathbb{C} I_n$.

Exercise 4.4 (A zero Killing form).

(i) Let $\mathfrak{g}$ be nilpotent, with $C^0\mathfrak{g} = \mathfrak{g}$ and $C^{k+1}\mathfrak{g} = [\mathfrak{g}, C^k\mathfrak{g}]$. Prove that $\kappa_{\mathfrak{g}} = 0$.

(ii) In $\mathfrak{gl}_3(\mathbb{C})$, let $t = \operatorname{diag}(1, i, 0)$, $x = E_{13}$ and $y = E_{23}$. Prove that their span $\mathfrak{g}$ is a subalgebra with $\kappa_{\mathfrak{g}} = 0$, and that $\mathfrak{g}$ is not nilpotent.

(iii) Explain what (ii) shows about Cartan's criterion.

Exercise 4.5 (Real subspaces of $\mathfrak{sl}_2$). Use the Gram matrix of the Killing form of $\mathfrak{sl}_2$ from the lecture.

(i) Prove that κ is positive definite on the real span of h, $e + f$ and $i(e - f)$.

(ii) Prove that κ has signature $(2, 1)$ on the real span of h, e, f, and find a nonzero vector x there with $\kappa(x, x) = 0$ (an isotropic vector).

(iii) Show that κ restricted to the subalgebra $\operatorname{span}\{h, e\}$ is degenerate.

Exercise 4.6 (Nilpotent ideals and the radical of κ). Let $\mathfrak{g}$ be any Lie algebra, and let $\mathfrak{g}^{\perp}$ be the radical of $\kappa_{\mathfrak{g}}$.

(i) Let $\mathfrak{n}$ be a nilpotent ideal. Prove that $\mathfrak{n} \subseteq \mathfrak{g}^{\perp}$.

(ii) Prove that $\mathfrak{g}^{\perp} \subseteq \operatorname{Rad} \mathfrak{g}$.

(iii) For the upper triangular algebra $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$, show that every nilpotent ideal lies in $\mathbb{C}E_{12} \oplus \mathbb{C}I_2$, and that the inclusion in (ii) is strict.